

A Projective Informational Framework

Why laws, particles, and time exist at all

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(Independent research framework)

1. Motivation: the problem behind the problems

Modern physics is extraordinarily successful, yet conceptually fragmented.

- Quantum Mechanics: probabilistic, nonlocal, measurement-dependent
- General Relativity: geometric, deterministic, local
- Cosmology: requires dark components with unclear ontology

But a deeper question remains largely untouched:

**Why does the universe admit stable laws, objects,
and a time direction at all?**

The MPFM does not aim to replace existing theories.

It addresses a prior structural question.

2. Core idea (one slide, one sentence)

Physical reality is what survives repeated non-faithful projections of information.

Everything else follows.

3. What is meant by “information”?

- Not Shannon bits
- Not computation
- Not observer-dependent

Information here means:

relational structure capable of being distinguished.

No discreteness is assumed.

4. Projection: the key mechanism

A projection is any mapping from total information to observable information that is:

- non-injective
- irreversible

- selective

Examples:

- measurement
- interaction
- communication
- coarse-graining

Projection is not optional:

it is unavoidable as soon as relations exist.

5. Consequence 1: emergence of time

- Laws are mostly time-reversal invariant
- Yet time has a clear arrow

In MPFM:

- time is not fundamental
- it emerges from accumulated irreversible information loss

Time = ordered memory of what cannot be recovered

6. Consequence 2: stable structures

Most relations are fragile under projection.

Only relations that are:

- symmetric
- topological
- phase-based
- redundant

survive repeated projection.

These survivors become:

- particles
- charges
- spins
- conservation laws

Reality is what resists erasure.

7. Particles as informational attractors

Particles are not fundamental objects.

They are:

informational attractors stable under projection

- Fermions: phase-based, highly robust
- Bosons: relational quanta of communication

This explains:

- universality of particles
- identical properties everywhere
- small number of stable species

8. Gravity: not a force, not entropic

In MPFM:

- mass = cumulative informational cost
- gravity = macroscopic visibility of accumulated projection deficit

Consequences:

- always attractive
- negligible microscopically
- dominant at large scales

No modification of GR equations is proposed.

9. Black holes (structural necessity)

A black hole appears when:

local informational deficit exceeds the projective capacity of the boundary

This yields:

- horizons
- area laws
- Bekenstein–Hawking entropy

Black holes are not exotic:

they are inevitable saturation states.

10. Cosmology and Λ

The cosmological constant is interpreted as:

a residual imprint of pre-projective information.

Key points:

- Λ is independent of matter
- linked to an informational scale ℓ_i
- no dynamical dark energy assumed

11. Life and consciousness (optional advanced section)

Life and consciousness are not new substances.

They are:

extremely robust informational attractors under noise and projection.

Consciousness corresponds to:

- reflexive informational loops
- high resistance index $\mathcal{R}$
- temporal integration

No mind-dependent physics is introduced.

12. What this framework is NOT

- not a replacement for QM or GR
- not a theory of everything
- not numerology
- not metaphysics

It is a structural layer, not a dynamical one.

13. Testability and limits

The model is falsifiable in principle via:

- spectral dimensionality ($D_s < 3$)
- information-loss scaling
- subtle interferometric effects
- large-scale structure correlations

It explicitly accepts:

possible future refutation.

14. Why this framework may be useful

It provides:

- conceptual unification without modifying equations
- clarity on why laws exist
- a coherent interpretation of time, gravity, and structure
- compatibility with all established physics

15. Final statement

Physics explains how the universe behaves.

This framework explains why a universe capable of laws exists at all.